DSA

Practical No. 5

Stack

Name: Arjun Mankuskar

PRN:202401110038

Branch: CSE (AIML)

Batch: A2

```
#include<iostream>
using namespace std;

class Node{
public:
   int data;
   Node* next;
   Node(int a){
      data = a;
      next = NULL;
   }
};

class Stack{
   Node* head;
public:
   Stack(){
      head = NULL;
   }
   void push(int a){
      Node* x = new Node(a);
      if(head == NULL){
         head = x;
      } else{
```

```
        x->next = head;
        head = x;
    }
}

void pop(){
    if(head == NULL){
        cout << "Empty" << endl;
        return;
    }
    Node* x = head;
    head = head->next;
    cout << "Popped: " << x->data << endl;
    delete x;
}

void display(){
    if(head == NULL){
        cout << "Empty" << endl;
        return;
    }

    Node* cur = head;
    while(cur != NULL){
```

```
            cout << cur->data;
            if(cur->next != NULL) ;
            cur = cur->next;
        }
        cout << endl;
    }
};

int main(){
    int ch;
    Stack s;
    do{
        cout << "1. push" << endl;
        cout << "2. pop" << endl;
        cout << "3. display" << endl;
        cout << "4. Exit" << endl;
        cout << "Enter your Choice: ";
        cin >> ch;
        switch(ch){
            case 1: {
                int a;
                cout << "Enter Value: ";
                cin >> a;
                s.push(a);
```

```
            break;
        }
        case 2:
            s.pop();
            break;
        case 3:
            s.display();
            break;
        case 4:
            cout << "Exiting" << endl;
            break;
        default:
            cout << "wrong choice" << endl;
      }
   } while(ch != 4);
   return 0;
}

/*
OUTPUT:-
--- Stack Menu ---
1. Push
2. Pop
3. Peek
```

4. Display

5. Exit

Enter your choice: 1

Enter value to push: 78

78 pushed into stack.

--- Stack Menu ---

1. Push

2. Pop

3. Peek

4. Display

5. Exit

Enter your choice: 1

Enter value to push: 96

96 pushed into stack.

--- Stack Menu ---

1. Push

2. Pop

3. Peek

4. Display

5. Exit

Enter your choice: 4

Stack elements (top to bottom): 96 78

--- Stack Menu ---

1. Push

2. Pop

3. Peek

4. Display

5. Exit

Enter your choice: 2

96 popped from stack.

--- Stack Menu ---

1. Push

2. Pop

3. Peek

4. Display

5. Exit

Enter your choice: 3

Top element: 78

--- Stack Menu ---

1. Push

2. Pop

3. Peek

4. Display

5. Exit

Enter your choice: 5

*/